KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2018
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is Machine Learning? Briefly discuss some practical applications of Machine Learning. (05)
   b) Explain McCulloch-Pitts model of artificial neuron with appropriate sketch. Describe significance of its individual components. (08)
   c) Explain the difficulties of solving XOR problem with single neuron. Solve XOR problem through manipulation by hidden neurons for AND and OR logic. (10)
   d) How does the cross-validation technique evaluate machine learning algorithms? Explain. (06)
   e) "The choices of training data and target concepts are two important issues in designing a machine learning system"– justify the statement with appropriate examples. (06)

2. a) What is the key difference between traditional algorithms and machine learning algorithms? (05)
   b) Write down the back propagation (BP) weight update rule for both hidden layer neurons and output layer neurons where sigmoidal and tanh functions are used as activation function in output and hidden layers, respectively. (10)
   c) How momentum parameter speed up training in BP? (06)
   d) What are MLP and RBF network? Write down the differences between MLP and RBF network. (06)
   e) Draw sketch diagram of a RBF network and put parameter values to solve XOR problem. (08)

3. a) Why ensemble of several neural networks is necessary when a network with hidden layer may approximate any arbitrary function? (05)
   b) What do you mean by independent, sequential and simultaneous training of component networks in ensemble construction? Mention prominent ensemble methods with each type of component network training. (10)
   c) Describe Bagging and AdaBoost ensemble methods with graphical representation. Compare the methods in different issues such as, training set size, ensemble decision and so on. (10)
   d) Describe an ensemble method which promotes diversity altering error of individual network. Is such technique suitable for ensemble of Decision Tree? Explain. (10)

4. a) What do you understand by Swarm Intelligence (SI)? Write down common features of SI based algorithms. (05)
   b) What do you mean by exploration and exploitation? Explain exploration and exploitation tradeoff in Particle Swarm Optimization (PSO). (08)
   c) "In PSO, each and every particle changes position in each iteration even the particle is the global best (G)"– are you agree with the statement? Justify your answer with appropriate assumptions in velocity update equation. (08)
   d) Write down the equations of transition probability and pheromone update of Ant Colony Optimization (ACO). Describe effects of key parameters in ACO. (09)
   e) Describe significance properties of ACO with respect to other population based algorithms. (05)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define GA. Explain the working principle of GA with a flowchart. (07)
   b) "Crossover is explorative and Mutation is exploitative"– justify the statement with respect to GA. What are the effects of too high and too low Mutation rate? (10)
   c) Write down the similarities and dissimilarities between GA and PSO. (06)
   d) Apply the Enhanced Edge Recombination algorithm on the following two chromosomes to produce two child. Show all steps. (12)

Parent 1:	4	2	1	3	5	6
Parent 2:	4	3	1	2	6	5

6. a) What is the problem of using normal crossover and mutation operators in GA for solving Travelling Salesman Problem (TSP)? Briefly explain Insert Mutation operator for TSP. (
   b) What is Decision Tree? Write down the steps of Decision Tree construction. (07)
   c) What is Information Gain? How does Information Gain handle continuous-valued attributes? (07)
   d) Consider the following table where attribute 'Status' represents class label. Find the root attribute of the Decision Tree using Gain Ratio as the attribute selection measure. (12)

Department	Age	Status	Salary
Sales	31··35	Senior	Medium
Sales	26··30	Junior	Low
Marketing	36··50	Senior	High
Secretary	26··30	Senior	Medium
Secretary	36··50	Senior	High
Sales	36··50	Junior	High
Marketing	31··35	Senior	Medium
Marketing	26··30	Junior	Low
Secretary	31··35	Junior	Medium
Secretary	26··30	Junior	Low

7. a) Distinguish between Supervised and Unsupervised learning. (06)
   b) What is the problem associated with K-means clustering algorithm? How K-mediod clustering method solve this problem? Explain. (12)
   c) What are the advantages of Hierarchical clustering over Partitioning method of clustering? (05)
   d) Consider the following 5 data points. (12)

   (1, 1), (3, 2), (7, 9), (8, 7), (9, 8)

   Find the clusters using Agglomerative clustering and use Complete Link proximity as distance measure. Show all steps.

8. a) What is the problem of DBSCAN algorithm? Discuss. (05)
   b) Explain core-distance and reachability-distance with graphical representation. (06)
   c) Briefly explain the curse of dimensionality problem. (06)
   d) Consider the following dataset with 2 dimensions. (18)

Dimensions 1	Dimensions 2
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
2.5	3.0
2.7	2.5
2.0	1.6
1.0	1.1
2.5	2.6
2.1	1.9

Using Principal Component Analysis (PCA), reduce the dataset to one dimension. Explain each step with necessary calculations.